



SDEV2171

Lesson 02: React Native and Expo Tooling Setup

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Lesson Objectives

By the end of this lesson, students should be able to:

- Install and configure the React Native development environment using Expo
- Verify required tooling and dependencies for mobile application development
- Set up and test an emulator or physical device for running Expo applications
- Identify and resolve common setup and configuration issues

Lesson Roadmap

1. Why the environment matters
2. Required tools and workflow choices
3. Instructor verification demo
4. Guided setup and verification
5. Troubleshooting and status tracking
6. Checkpoint and next steps

What Success Looks Like Today

- Your machine can start an Expo project
- You can open that app on an emulator or physical device
- You can identify what is still missing if setup is incomplete
- You leave with one of three states:
 - `verified`
 - `partial`
 - `blocked`

Why This Environment Matters

- It is the foundation for every coding lesson that follows
- It supports fast feedback while students learn UI and interaction concepts
- It reduces friction when testing on real mobile devices
- It creates a repeatable development workflow for assignments

Tooling Overview

- Node.js runtime
- `create-expo-app` for creating projects
- Expo CLI workflow through `npx expo ...`
- development build and Expo Go runtime choices
- Android emulator and/or physical device workflow
- Code editor with terminal access

Course baseline

- Expo SDK 56
- React Native 0.85
- React 19.2.x
- Node.js 22.13.x or newer

Two Runtime Paths

Expo Go

- a shared prebuilt Expo runtime
- useful for understanding the simplest Expo launch idea
- limited when the course needs newer native setup or stronger DevTools behaviour

Development build

- your own installed app built from the project
- closer to the real app runtime the course uses later
- the primary course path after setup is complete

Course Runtime Rule

- show both Expo Go and development builds so the difference is visible
- use development builds as the primary course workflow
- keep Expo Go as a comparison or fallback concept, not the main class path

Why

- later lessons depend on React Native DevTools more directly
- current Expo and React Native tooling is more reliable on the development-build path

Version Note

- Official Expo docs checked on 2026-06-04T04:16Z
- Expo SDK 56 is the current course standard
- SDK 56 uses React Native 0.85 and React 19.2.3
- Expo SDK 56 requires Node.js 22.13.x or newer
- This course uses the provided verification app to keep lesson 02 on one shared baseline
- Project creation choices are handled separately in lesson 03 materials because create-expo-app@latest without an SDK-specific template still creates SDK 54 during the current transition period

Testing Paths

Emulator path

- useful when you do not want to rely on a phone
- good for repeatable testing during class
- best shared classroom path for the course development build

Physical-device path

- useful only if the SDK 56 device workflow has already been prepared
- can be used with a prepared development build or a prepared Expo Go comparison path
- otherwise, stay on emulator or simulator for the shared class path

Course rule

At least one path should be verified today.

Shared Setup Sequence

1. Confirm terminal and Node.js access
2. Confirm editor and project folder workflow
3. Confirm device path:
 - emulator ready, or
 - physical device ready
4. Open the provided Expo verification app
5. Add dev-client support
6. Build and install the development client on one path
7. Start the Expo server
8. Run the app and observe the verification screen

Exact startup path

- From the repo root, open:
 - `resources/01-foundations-setup/lesson-02/example/environment-check-app`



Verification Demo

Watch for these checkpoints

- Terminal commands run without path errors
- development client build installs successfully
- Expo server starts successfully
- launcher flow or automatic app open appears
- App opens on an emulator or device
- Visible confirmation screen loads

What To Do If You Are Blocked

- Check the current step before trying random fixes
- Use the troubleshooting checklist
- Record your status as `partial` or `blocked`
- Use an alternate verification route if needed:
 - pair with instructor demo
 - a prepared shared device path
 - defer one setup step while confirming the rest

Evidence to capture

- failing step number
- exact error or symptom
- first action tried
- result after retry

Setup Status Tracker

verified

- Verification app runs on your chosen device path

partial

- Some tools work, but one major step remains

blocked

- You cannot complete the current path without additional help or access

Instructor completes

- validation of current state
- approved next action for lesson 03

Guided Practice Priorities

- Follow the shared checklist in order
- Verify one working device path before optimizing anything else
- Avoid deep customization
- Ask for help with concrete evidence:
 - current step
 - exact error
 - verified vs partial vs blocked

Common Blockers

- `node` or `npm` command not found
- Emulator not installed or not starting
- Device path not prepared for SDK 56
- Expo server starts but app does not connect
- App does not refresh after a code change

Response pattern

Identify the failing step, then test the smallest next action.

Checkpoint

Students should be ready to show one of these

- a running verification app on emulator or device
- documented progress plus the next required action

Students should also be able to answer

- Which testing path are you using?
- What is your current status?
- What is your next step if you are not yet verified?

What We Learned Today

- Which tools and versions make up the course baseline
- Why at least one working device path must be verified early
- How to start and verify the shared Expo app
- How to classify progress as `verified`, `partial`, or `blocked`
- How to capture evidence and take the next troubleshooting step

Next Lesson Bridge

- If you are **verified** , you are ready to create your own Expo app in lesson 03
- If you are **partial** , bring your status tracker and continue from the recorded next step
- If you are **blocked** , bring the recorded blocker so class support starts from evidence, not memory